

BUSINESS INTELLIGENCE AND TOOLS

(24CSH-341)

EXPERIMENT 3: Dimensional Modeling and Star Schema in Tableau

AIM To design and implement a dimensional data model using the Tableau ecosystem by transforming transactional data from the Global Superstore dataset into a Star Schema. This schema, consisting of central Fact and surrounding Dimension tables, will be built to ensure highly efficient reporting and analytical processing.

OBJECTIVES

- To thoroughly understand the concepts of Dimensional Modeling and the Star Schema architecture.
 - To clearly differentiate between Fact Tables (metrics) and Dimension Tables (descriptive attributes).
 - To transform flat, transactional data into a structured dimensional model using Tableau Prep Builder and Tableau Desktop.
 - To establish the correct logical relationships between the newly created Fact and Dimension tables.
 - To create analytical measures (using Tableau Calculated Fields in place of DAX) for comprehensive reporting.
 - To validate the integrity of the Star Schema using Tableau visualizations.
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TOOLS & SOFTWARE REQUIRED

- Tableau Desktop
 - Tableau Prep Builder (for creating the distinct Fact and Dimension tables)
 - **Dataset Used:** [Global Superstore Dataset.xlsx](#)
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THEORY

The Architecture of Dimensional Modeling Dimensional Modeling is a specific data warehouse design technique heavily used in Business Intelligence systems. Its primary purpose is to organize and structure data specifically for analytical processing and reporting.

To build this, we split our data into two very distinct types of tables:

- **Fact Tables:** These act as the core of the model. They contain measurable business events and numerical values. In our scenario, this includes quantitative data such as Sales, Profit, Quantity, and Discount.
- **Dimension Tables:** These provide the context. They contain the descriptive attributes used to analyze and slice the facts. Examples from our dataset include Customer, Product, Region, and Date details.

The Star Schema A Star Schema is a specific type of dimensional model where one central Fact Table is directly connected to multiple surrounding Dimension Tables. This design looks like a star, hence the name. It simplifies reporting, drastically improves analytical query performance, and provides a highly structured foundation for BI analysis.

In this experiment, the raw transactional Orders data is transformed into a central [Fact_Sales](#) table, while the descriptive information is organized into separate dimension tables: [Dim_Customer](#), [Dim_Product](#), [Dim_Region](#), and [Dim_Date](#).

PROCEDURE

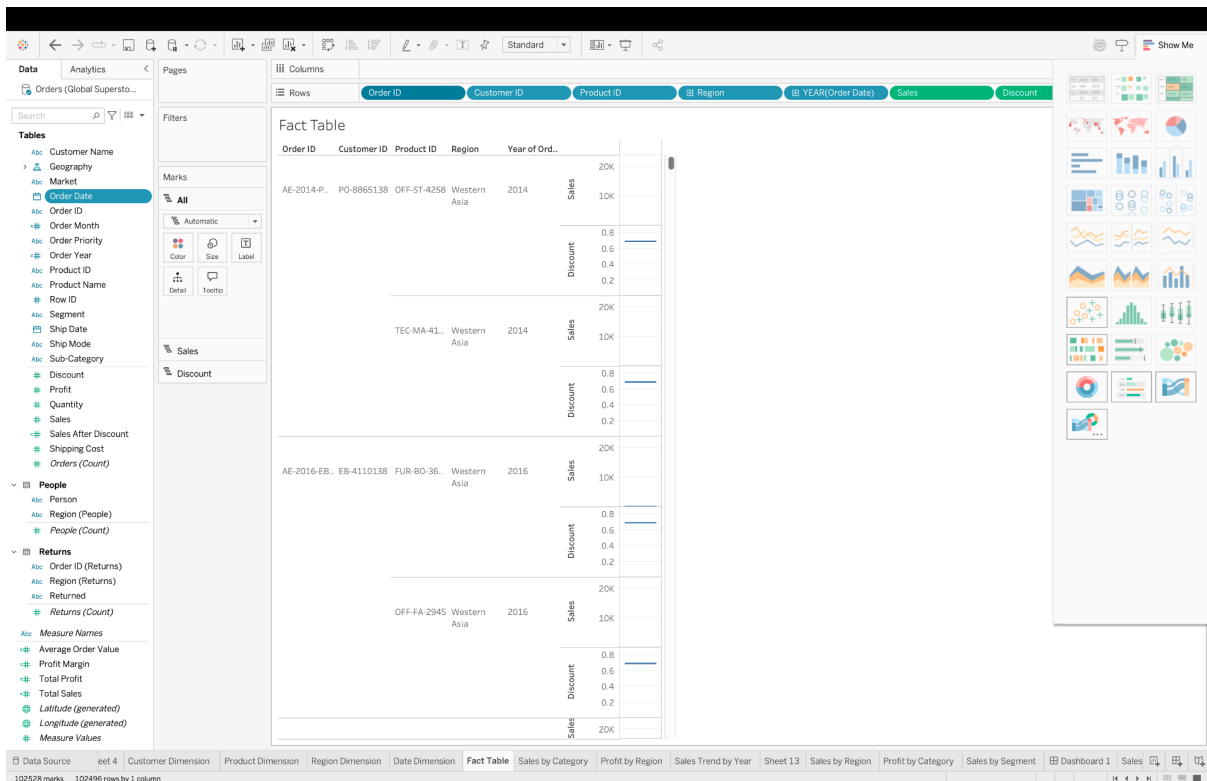
Part A: Load the Dataset

- **Step 1: Open Tableau**
 - Launch Tableau Prep Builder (to build the physical tables) or Tableau Desktop.
- **Step 2: Import the Global Superstore Dataset**
 - Connect to a file and select the **Global Superstore Dataset.xlsx** workbook.
- **Step 3: Select the Required Worksheets**
 - Import the **Orders**, **Returns**, and **People** worksheets into your environment.
 - Verify that all imported tables appear correctly.

Part B: Identify Fact and Dimension Attributes

- **Step 4: Analyze the Orders Table for Facts**
 - Examine the Orders data to identify the measurable business attributes.
 - Isolate the numerical fields: Sales, Profit, Quantity, and Discount. These represent measurable business events and belong in the Fact Table.
- **Step 5: Identify Descriptive Attributes**
 - Identify the fields that describe the *who*, *what*, *where*, and *when* of the transactions.

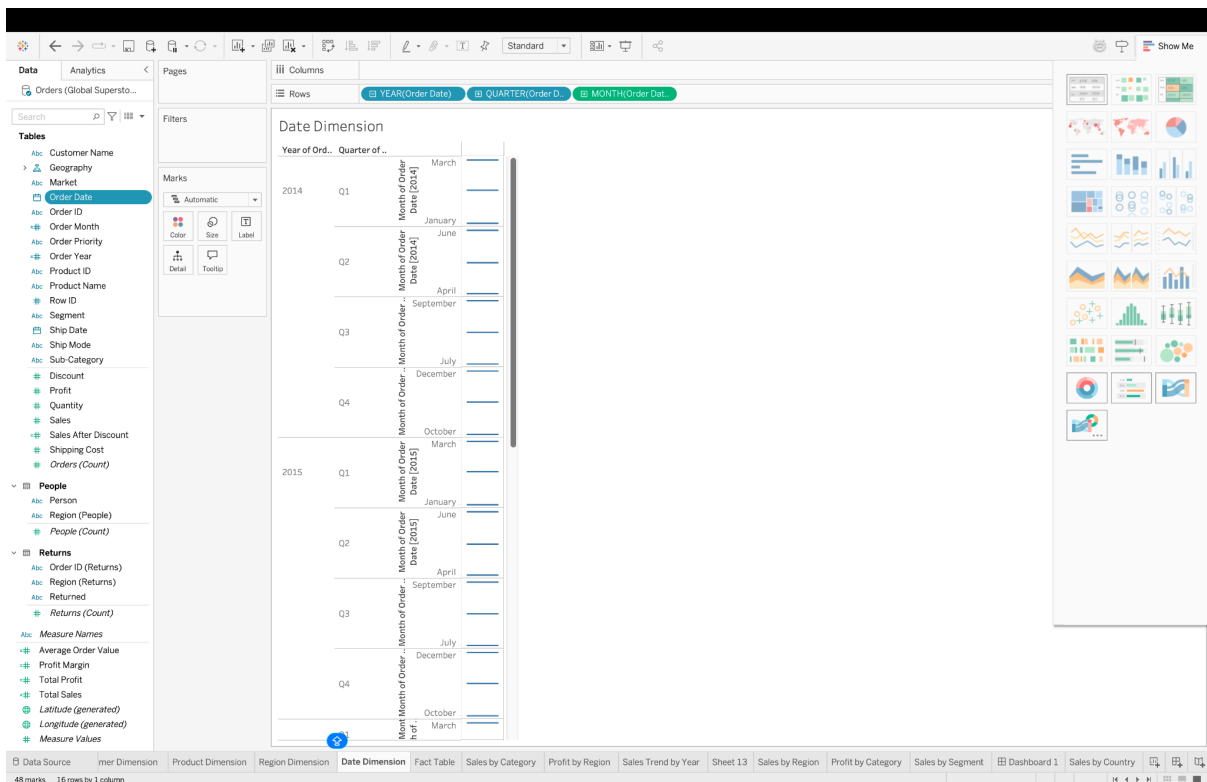
- Group these into categories: Customer, Product, Region, and Date.
- **Step 6: Categorize the Data Strategy**
 - Plan the Star Schema classification:
 - **Fact Table:** Fact_Sales
 - **Dimension Tables:** Dim_Customer, Dim_Product, Dim_Region, Dim_Date



Part C: Create the Dimension Tables (Using Tableau Prep)

- **Step 7: Start the Transformation**
 - Open your workflow to begin transforming the raw data into separate outputs.
- **Step 8: Create Dim_Customer**
 - Branch off a step containing only: Customer ID, Customer Name, and Segment.
 - Use the aggregate/grouping function to remove all duplicate records so each Customer ID appears only once.
- **Step 9: Create Dim_Product**
 - Branch off a step containing: Product ID, Product Name, Category, and Sub-Category.

- Remove duplicate records to ensure uniqueness.
- **Step 10: Create Dim_Region**
 - Branch off a step containing: Region, Country, State, and City.
 - Remove duplicate records.
- **Step 11: Create Dim_Date**
 - Branch off a step containing: Order Date, Year, Quarter, and Month.
 - Ensure all date-related fields are correctly formatted.
- **Step 12: Verify Unique Keys**
 - Double-check that duplicate records have been removed from all dimension tables and that the primary key fields contain strictly unique values.



Part D: Create the Fact Table

- **Step 13: Create Fact_Sales**
 - Branch off a final step to act as the central Fact Table. It must contain the foreign keys and measures: Order ID, Customer ID, Product ID, Region, Order Date, Sales, Quantity, Profit, and Discount.
- **Step 14: Verify Key Columns**
 - Ensure all key columns needed to establish relationships with the Dimension Tables are present and formatted correctly.

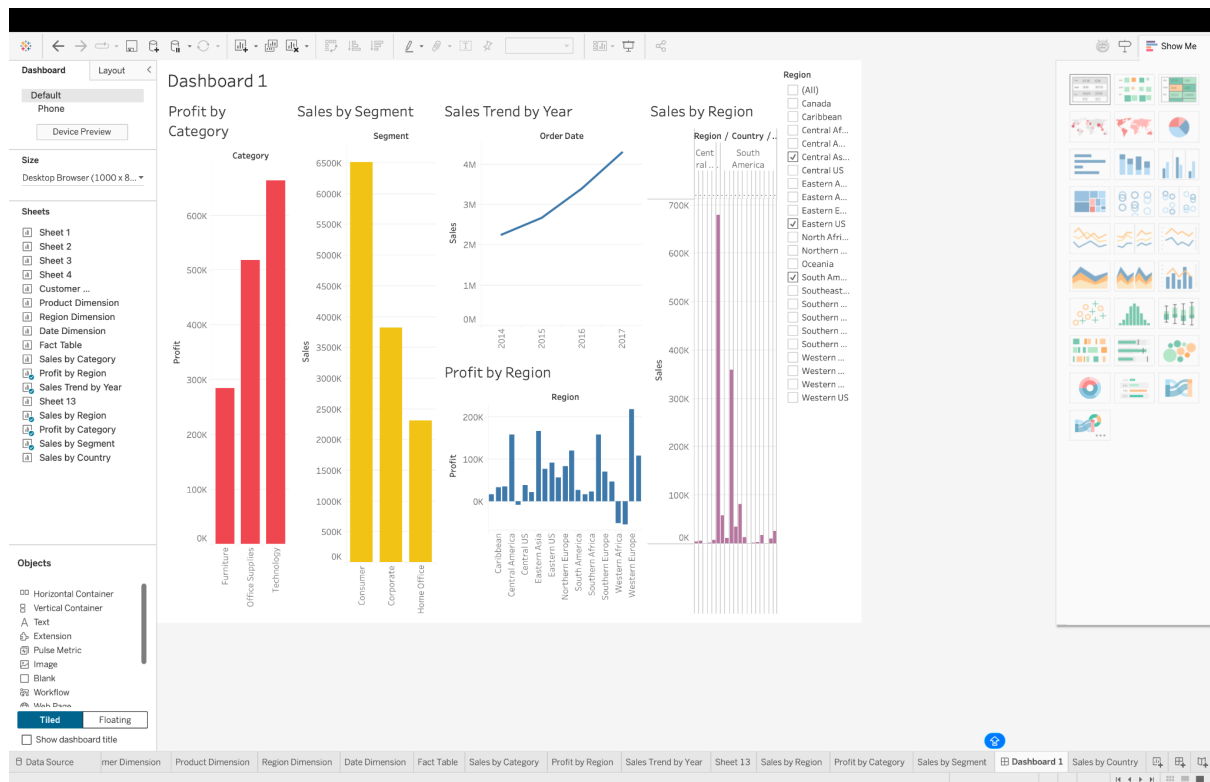
Part E: Build the Star Schema (Tableau Desktop)

- **Step 15: Open the Data Model Canvas**
 - In Tableau Desktop, navigate to the Data Source page (the logical layer canvas).
- **Step 16: Establish Relationships (Noodles)**

- Drag **Fact_Sales** to the center of the canvas.
- Create logical relationships connecting **Fact_Sales** to **Dim_Customer**, **Dim_Product**, **Dim_Region**, and **Dim_Date** using their respective ID fields.
- Configure these relationships as Many-to-One, ensuring the Fact Table is on the "many" side and the Dimension Tables are on the "one" side.
- **Step 17: Arrange the Model Visually**
 - Organize the tables on the canvas so that **Fact_Sales** sits centrally, with the four dimension tables branching out, perfectly forming the Star Schema.

Part F: Validate the Model and Create Analytical Measures

- **Step 18: Create Calculated Fields (Tableau's equivalent to DAX)**
 - Navigate to a new worksheet and create the following analytical measures:
 - **Total Sales:** $SUM([Sales])$
 - **Total Profit:** $SUM([Profit])$
 - **Profit Margin:** $SUM([Profit]) / SUM([Sales])$
 - **Average Order Value:** $[Total Sales] / COUNTD([Order ID])$
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- **Step 19: Create Validating Visualizations**
 - Build visualizations using the dimensional model to analyze the data.
 - Create a **Sales by Category** chart, a **Profit by Region** map, and a **Sales Trend by Year** line graph.
- **Step 20: Validate the Star Schema Integrity**
 - Use fields from the newly created Dimension Tables as filters on the Fact Table.
 - Verify that clicking or filtering dimensions (Customer, Product, Region, Date) correctly affects the Fact Table data and updates the dashboard visualizations accurately.



RESULT

The raw data was successfully extracted from the Global Superstore dataset and transformed into a highly efficient dimensional model. Using data preparation techniques, centralized Fact tables and unique Dimension tables were generated to form a strict Star Schema. Data cleaning, standardization, logical relationship creation, and metric generation were performed successfully, resulting in a dataset fully optimized for advanced reporting and business analysis.

LEARNING OUTCOMES

- Understood the ETL process and its role in Business Intelligence.
- Learned how to extract data from a real-world business dataset.
- Gained practical experience with data preparation and transformation tools.
- Learned how to clean, filter, transform, and successfully merge datasets.
- Learned how to handle duplicate records, blank values, and correct data type inconsistencies.

- Learned how to create robust logical relationships between multiple tables in a BI environment.
- Learned how to create powerful analytical measures for calculations.
- Understood the practical application and business use of Total Sales, Total Profit, Profit Margin, and Total Orders metrics.
- Learned how to author BI visualizations and interactive dashboards for comprehensive business analysis.
- Gained practical experience in architecting and preparing a dimensional data model for enterprise reporting and decision-making.